

SCC 451 Machine Learning Final Report

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Abstract— This report evaluates the efficacy of distinct machine learning methodologies applied to both low-dimensional meteorological data and high-dimensional image datasets. The study implements a comprehensive data mining pipeline following the CRISP-DM framework, encompassing data pre-processing, dimensionality reduction via Principal Component Analysis (PCA), and unsupervised clustering using K-Means, DBSCAN, and Gaussian Mixture Models (GMM). Results indicate that K-Means (K=3) offers the most robust segmentation for the Basel Climate dataset, effectively capturing seasonal variations when coupled with PCA. For the image processing task, Deep Neural Networks (ResNet-50) were utilized for feature extraction on the *OxfordPets* and *Cats vs Dogs* datasets. The analysis demonstrates that linear classifiers and Support Vector Machines (SVMs) achieve high predictive accuracy (up to 99%) on deep feature embeddings, while unsupervised clustering successfully discover the underlying class structures even in the absence of labels.

I. INTRODUCTION

Machine Learning is a data-driven paradigm and a subset of artificial intelligence that focuses on learning patterns in data and making inferences about new data. It is used in a variety of tasks – including but not limited to classification, clustering, predictions and is even used in fields like computer vision. However, machine learning isn't simply limited to training models on data but rather encompasses a variety of tasks from data understanding, visualization and pre-processing to building entire predictive pipelines. It provides the backbone of most modern AI systems, from forecasting models to autonomous vehicles to large language models (LLMs) and other generative AI tools. [1]

We cannot cover them all in this paper, however, the following section aims to provide a basic overview of some of the most commonly used techniques.

When building a machine learning pipeline, several steps are involved. Many industrial frameworks have been established, however, the most commonly used one is known as the CRISP-DM(Cross-Industry Standard Process for Data Mining) framework.

The CRISP-DM methodology breaks down the life cycle of a data mining project into **six major phases** [Figure 1]. These phases are designed to move from high-level objectives to concrete results, allowing for frequent iteration and refinement.

A. Business and Data Understanding

This phase focuses on clearly defining the project objectives and aligning them with appropriate machine learning goals. It involves identifying the problem type (such as classification, regression, or clustering) and translating high-level business or academic aims into measurable success criteria. In parallel, initial data collection and exploratory data analysis (EDA) are performed to understand the size, structure, and characteristics of the data, identify quality issues such as missing values or outliers, and explore key patterns or

correlations. The insights gained at this stage inform early hypotheses and help shape a realistic and well-aligned project plan.

B. Data Preparation/Pre-Processing

The first step to building any machine learning pipeline is to pre-process the data in a way that is not only understandable for the model but also does not introduce any bias or discrimination while training. It is an essential step since it directly impacts the performance and accuracy of the model.

It involves several techniques such as cleaning the data to handle missing values, removing outliers, scaling features, encoding categorical variables, and splitting the data into training and testing sets. These techniques are key for ensuring the data is in a consistent and usable format for the ML algorithms.[2]

C. Dimensionality Reduction (Data Preparation/Pre-Processing)

Dimensionality reduction is a subset of pre-processing that aims to reduce the number of features (variables) in a dataset while also not losing the important features that may affect the accuracy and prediction of data. It is especially important for distance-based techniques like clustering where high-dimensionality can make it difficult for models to infer distances between features in a feature space since it may make the data sparse and models inefficient.

Some commonly used techniques are feature selection where domain knowledge is used to pick the most important features. Others include PCA or distance based metrics like T-SNE or UMAP which try to capture the variance of the dataset with reduced dimensionality.

These techniques not only help reduce noise and combat high-dimensionality but also help in visualizing the data easily.

D. Modeling

Machine learning models are programs that learn patterns from data and use them to make predictions or decisions on new, unseen data. Most machine learning approaches fall into three main categories: **supervised learning**, **unsupervised learning**, and **reinforcement learning**.

In **supervised learning**, the model is trained on labeled data, meaning the correct answers are already known. The goal is to learn the relationship between input and outputs. Common tasks include **classification** (e.g., image recognition using models like Logistic Regression or Convolutional Neural Networks) and **regression** (e.g., predicting population growth or health outcomes using Linear Regression or Random Forests). [3]

In **unsupervised learning**, the model is given data without labels and learns to find patterns or group similar data points on its own. A common task is clustering, which is used in recommendation systems on online stores.

Typical models include **K-Means** and **Hierarchical Clustering**. [3]

In **reinforcement learning**, the model learns by interacting with an environment and improving through trial and error, guided by rewards or penalties. This approach is often used in robotics and autonomous driving, with models such as **Q-Learning** and **Deep Q-Networks (DQN)**. [3]

E. Evaluation

Model evaluation in machine learning is the process of measuring how well a model performs its intended task. It allows practitioners to compare different models or training runs and determine which approach is most effective for a given problem.

The evaluation metrics used depend on the type of task. For **classification**, common metrics include **accuracy**, **precision**, **log-loss**, **AUC**, and the **confusion matrix**. For **regression**, typical metrics are **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and **Mean Absolute Error (MAE)**.

While many evaluation metrics exist across different domains, these are the most commonly used ones. This task is linked to **Business Understanding** because evaluation metrics must reflect the project's objectives. Aligning metrics with business objectives ensures that model performance is meaningful and actionable.

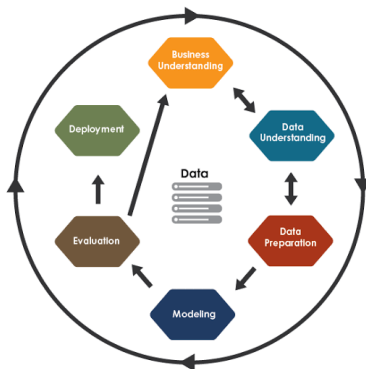


Figure 1. CRISP-DM framework.

II. LITERATURE REVIEW

When comparing the methodologies and what models to use for the tasks, we delved into the literature to understand the advantages and pitfalls of each model.

A. Clustering

Clustering is an unsupervised learning technique that groups data points based on similarities or patterns [4]. A good cluster maximizes the distance between clusters while minimizing the distance within a cluster. Clustering algorithms can be categorized in several ways, such as **hard vs. soft clustering** or based on their strategy, including **partitioning**, **hierarchical** and **density-based** methods. [4]

For the two tasks, we used multiple clustering methods and compared their performances against each other. We started with a simple base clustering algorithm with K-means and then moved on to a Gaussian mixture model and a density based DB-scan to evaluate the performance. The following provides a brief summary of each model.

a) *K-Means Clustering*: The most popular clustering method, K-Means, is a partition-based algorithm that begins by selecting K centroids at random. Each data point is then assigned to its nearest centroid, and the centroids are updated iteratively until a convergence criterion, such as the square-error, is met. Despite its popularity, K-Means is sensitive to noise and outliers, which can significantly affect the centroids if not properly handled. It is also prone to poor initialization, as bad starting points can cause the algorithm to get trapped in a local optimum.[4]

b) *DBSCAN*: DBScan is a density-based clustering algorithm that groups data points based on two parameters: the neighborhood radius and the minimum number of points within that radius. Points on the edge of a neighborhood define cluster boundaries, while points that do not belong to any neighborhood are classified as noise. This makes DBScan well-suited for datasets with noise, outliers, or arbitrarily shaped clusters. However, it is sensitive to high-dimensional data and can perform poorly in such spaces. Its performance also depends heavily on the choice of input parameters, namely the radius and minimum points (MinPts).[4]

c) *Gaussian Mixture Models*: A Gaussian Mixture Model tries to explain a dataset as a combination of several multi-dimensional bell-shaped distributions (Gaussians). You can think of it as an extension of K-Means that works in situations where k-means struggles, such as when clusters overlap or have different shapes. Unlike K-Means, which assigns each data point to exactly one cluster, a GMM uses soft clustering. This means each data point is assigned a probability of belonging to each cluster, rather than a single hard label. Behind the scenes, a GMM uses an iterative process called expectation-maximization (EM). [6]

B. Cluster Evaluation

Cluster evaluation or Cluster Validity Index(CVI) is a metric that is used to evaluate the quality of clustering results. While there are a wide range of cluster evaluation methods, in this work we focus on mainly two. The Silhouette Score and the Davies-Bouldin index.

The Silhouette score/index is a normalized summation-type index. It is measured based on the distance between all the points in the same cluster and the separation between clusters is based on the nearest neighbor distance. A higher silhouette index means a better performing cluster. While computationally expensive, it has the most consistent performance across varied scenarios and is best for measuring overlapping clusters and handling noise. [7]

The Davies-Bouldin (DB) index measures the cohesion of clusters based on the distance from the points in a cluster to its nearest centroid and the separation based on the distance between centroids. A lower DB index means better clustering. While it is one of the most commonly used indexes due to its computational efficiency, it struggles with complex shaped clusters and mostly favors spherical or globular clusters.[7]

Throughout this paper, when clustering we use the silhouette score as the primary index when evaluating clusters, with the Davies-Bouldin index being a close second.

C. ResNet50

ResNet (Residual Network) is a convolutional neural network (CNN) that uses skip connections to bypass layers, forming residual blocks. This design allows very deep networks to train effectively by mitigating the vanishing/exploding gradient problem.

ResNet50, developed by Microsoft Research in 2015, is a 50-layer deep CNN. Its architecture includes convolutional layers for feature extraction, identity and convolutional blocks that transform and process features through skip connections, and fully connected layers that perform the final classification. By stacking these residual blocks, ResNet50 can learn complex features without performance degradation in deep networks. [8]

III. TASK 1. BASEL CLIMATE DATASET

In this task, the data contains 1763 18-dimensional records from summer and the winter seasons in Basel, Switzerland. The goal is to cluster the data based on the seasons.

A. Data Pre-processing

Missing value analysis revealed the dataset contained no missing values.

The first observation was that the data had unmatched scales. Temperature values ranged from -5 to 30, while humidity ranged from 40 to 90 [Figure 2]. Similarly, wind gust ranged from 0 to 80, and sea level pressure ranged from 980 to 1050. Because of these unbalanced scales, clustering algorithms trained on the raw data would tend to favor variables with larger numeric ranges, potentially reducing accuracy. To address this, we implemented two scaling strategies. Standardization (z-score normalization) was applied by subtracting the mean and dividing by the standard deviation for each variable, while robust scaling, which subtracts the median and divides by the interquartile range, was applied to variables with skewness or outliers. Additionally, a $\log(1+x)$ transformation was used for variables exhibiting extreme skewness.

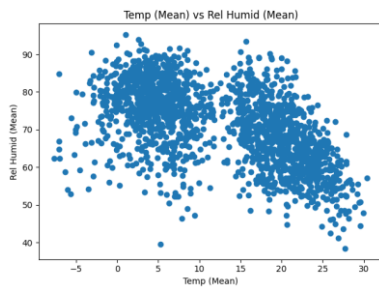


Figure 2 Temperature vs Humidity.

Heavy multicollinearity was also observed among similar variables such as temperature, wind gust, wind speed, relative humidity, and sea level pressure [Figure 7]. To reduce redundancy, we dropped the maximum and minimum feature columns for these variables and retained only the mean, as it sufficiently represented the underlying information.

B. Dimensionality Reduction

To address this, we applied Principal Component Analysis (PCA) to capture the variance and reduce the dimensionality of the data. we considered two approaches:

one using the preprocessed data without dimensionality reduction and another with PCA. By analyzing the scree plot, we found that four principal components were sufficient to capture the majority of the variance in the dataset [Figure 6].

C. Clustering

1) Kmeans:

We began with baseline clustering using KMeans, exploring cluster sizes from two to eleven. For the preprocessed data without dimensionality reduction, evaluation using the Scree plot, Silhouette Score, and Davies-Bouldin Index indicated that three clusters were optimal, achieving a Silhouette Score of **0.318** and a minimum Davies-Bouldin Index of **1.25**.

Similarly, for the dimensionality-reduced data, the same methods also suggested three clusters as optimal, with a slightly higher Silhouette Score of **0.35** while the Davies-Bouldin Index remained at **1.117**. In both cases, a cluster size of three provided the best balance between compactness and separation.

2) DBSCAN

For the preprocessed data without dimensionality reduction, the algorithm proved sensitive to parameter tuning. The optimal configuration was found at $\text{eps}=1.0$ and $\text{min_samples}=5$. This yielded 2 clusters with a Silhouette Score of **0.192** and a Davies-Bouldin Index of **2.203**. Lower eps values (e.g., 0.5) resulted in excessive fragmentation (17+ clusters, largely noise), while higher values merged all data into a single cluster.

The application of DBSCAN to the dimensionality-reduced data proved ineffective. Despite optimizing the epsilon parameter ($\text{eps}=0.5$), the algorithm fragmented the dataset into 10 clusters, which is inconsistent with the broader analysis. The resulting Silhouette Coefficient of **-0.20** indicates a significant degree of misclassification, suggesting that points were assigned to clusters where they did not belong. Furthermore, the high Davies-Bouldin Index (**3.134**) confirms poor separation between these groups. These metrics demonstrate that the density-based approach struggled to adapt to the topology of the reduced feature space, overfitting local noise rather than capturing the global structural patterns found by K-Means.

3) Gaussian Mixture Models:

Finally, Gaussian Mixture Models (GMM) were applied to accommodate the potential elliptical nature of weather clusters, which the spherical assumption of K-Means might misinterpret. we evaluated component counts ranging from 2 to 10 using the Bayesian Information Criterion (BIC) and Silhouette analysis.

On the pre-processed dataset without dimensionality reduction, the analysis suggested an optimal structure of 6 clusters (**BIC = -8281**). However, the Silhouette score was relatively low at **0.146**, while the Davies-Bouldin index was **2.408**.

For the dimensionality-reduced (PCA) data, the model converged on 4 clusters, yielding an improved Silhouette score of **0.257** and a BIC of **17,596**.

It is critical to note, however, that the Silhouette Coefficient is often an imperfect metric for evaluating GMMs. Silhouette penalizes overlapping clusters, favoring

distinct, separated boundaries. In contrast, GMM is a "soft" clustering method that probabilistically assigns data points to clusters, explicitly allowing for overlap. In the context of climate data, this property is theoretically advantageous: weather patterns often exist on a continuum rather than in discrete categories. For instance, transitional seasons like Autumn or Spring naturally lie between the extremes of Summer and Winter. Consequently, the "overlap" detected by the GMM likely reflects these genuine transitions rather than model failure. The figure below represents the clusters that were defined by GMM for 4 components.[Figure 3]

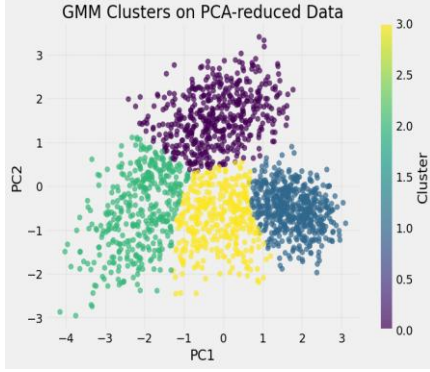


Figure 3 Clusters.

D. Conclusion

The following tables provide the final results of the clustering for both the datasets.

Algorithm	Silhouette Score	Optimal clusters	Davies Bouldin	BIC
K-Means	0.318	3	1.25	NA
DBSCAN	0.192	2	2.203	NA
Gaussian Model	0.146	6	2.408	-8281

Table 1 non-PCA pre-processed dataset results

Algorithm	Silhouette Score	Optimal clusters	Davies Bouldin	BIC
K-Means	0.35	4	1.117	NA
DBSCAN	-0.205	10	3.134	NA
Gaussian Model	0.257	4	1.379	17596

Table 2 PCA pre-processed dataset results

The analysis of the Basel Climate Dataset shows that both the choice of clustering algorithm and the way the data is represented play a crucial role in identifying weather patterns.

1) Impact of Dimensionality Reduction

One important finding is the benefit of using Principal Component Analysis (PCA). Clustering directly on the original scaled features produced only moderate separation of the data. However, when the data was projected onto its principal components, cluster cohesion improved noticeably. For example, K-Means' Silhouette score increased from 0.312 on the original data to 0.352 after PCA. This suggests that PCA effectively reduced noise

while keeping the main patterns that distinguish different weather regimes.

2) Algorithm Comparison

a) *K-Means*: This algorithm performed the best for this dataset. On the PCA-reduced data, it achieved the highest Silhouette score (0.352) with three clusters. This indicates that the climate data naturally organizes into three distinct states, likely reflecting seasonal or large-scale weather patterns.

b) *Gaussian Mixture Models (GMM)*: GMM gave a slightly different perspective. Its Silhouette scores were lower (0.286), but the Bayesian Information Criterion (BIC) suggested four clusters. This highlights that weather data often has transitional states—like autumn blending into spring—that GMM can capture more softly than K-Means, which assigns each point to a single cluster.

c) *DBSCAN*: This density-based method did not perform well, producing negative Silhouette scores (-0.20). The results suggest that climate data doesn't form clearly separated dense regions, making centroid-based methods like K-Means more suitable.

Overall, K-Means with three clusters on PCA-reduced data is the most reliable and interpretable choice for this dataset. Still, the GMM results remind us that weather patterns are connected by continuous transitions rather than rigid boundaries.

IV. TASK II: IMAGE PROCESSING WITH DEEP NEURAL NETWORKS

In this task, the goal was to extract feature vectors using Deep Neural Networks and then cluster and classify them based on different methods. Classification was supposed to be done on two datasets. we chose the [OxfordPetsIIT](#) and [Cats vs Dogs](#) dataset.

The Oxford-IIT Pet Dataset, is a fine-grained collection comprising 37 pet breeds, including both cats and dogs, with roughly 200 images per class, amounting to a total of approximately 7,349 images. This dataset poses a considerable challenge due to the high visual similarity between different breeds.

The second dataset, the Kaggle Cats vs Dogs Dataset, is a binary classification dataset containing 25,000 images, evenly split between cats and dogs (12,500 images per class). In contrast to the Oxford dataset, this collection focuses primarily on differentiating between the two species rather than distinguishing individual breeds.

A. Data Pre-processing

Pre-processing was standardized to align with the requirements of the ResNet-50 model using IMAGENET1K_V2 weights. The pipeline involved:

- **Resize**: Bilinear interpolation to 232×232 pixels.
- **Crop**: Central crop to 224×224 pixels.
- **Normalization**: Tensor values were scaled to the range $[0, 1]$ and standardized channel-wise using ImageNet means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225).

This process yielded tensors of shape (3, 224, 224), which were passed through the network to obtain the final 2048-dimensional feature embeddings.

Once the final 2048-dimensional features were extracted, Standardization (z-score normalization) was applied by subtracting the mean and dividing by the standard deviation. This was essential for dimensionality reduction so that one feature does not dominate the Principal Component Analysis.

B. Dimensionality Reduction

Given the high dimensionality ($d=2048$) and sparsity inherent in deep neural representations, we applied Principal Component Analysis (PCA) to project the feature vectors into a lower-dimensional latent space. This step was critical to mitigate the "Curse of Dimensionality," which can degrade the performance of distance-based clustering algorithms like K-Means. Rather than arbitrarily selecting a single hyperparameter, we conducted a comparative analysis by retaining the top 100, 50, and 20 principal components. This multi-scale approach allowed us to empirically evaluate the trade-off between information retention (explained variance) and the discriminative power of the resulting embeddings for both clustering and classification tasks.

A simple plot for the two ground truth classes in the Cats vs Dogs dataset is shown in [Figure 4 Image Labels]

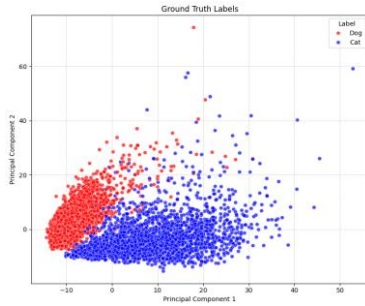


Figure 4 Image Labels

C. Clustering for Dataset (Cats vs Dogs)

we performed K-Means clustering on feature vectors reduced to 100, 50, and 20 principal components. The optimal validation metrics for each configuration are presented in [Table 3 Clustering Scores]. The results indicate that reducing the dimensionality to 20 components significantly enhances cluster definition, yielding the highest Silhouette Score (0.248) and the lowest Davies-Bouldin Index (1.829). This confirms that a compact feature representation (approx. 1% of the original embedding size) captures the essential class-separating variance while minimizing the adverse effects of high-dimensional sparsity.

Dataset Size	Clusters	Silhouette Score	Davies-Bouldin Index
100	2	0.169	2.611
50	2	0.195	2.269
20	2	0.247	1.828

Table 3 Clustering Scores

The following image shows a side by side comparison of the original labels versus the clustering labels.

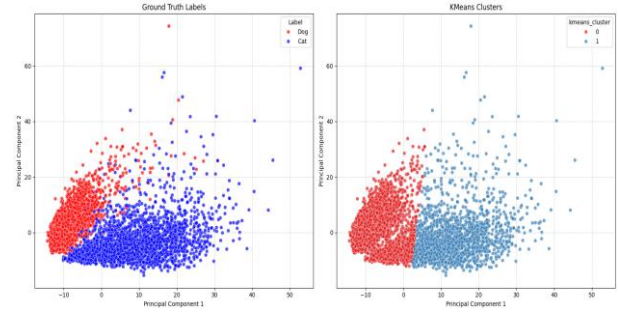


Figure 5 Original vs Predicted.

Visual inspection confirms that the classifier is predominantly accurate, with only sparse errors observed across the dataset.

D. Classification on Dataset (Cats vs Dogs)

For classification, we evaluated two methods: a support vector machine (SVM) and a linear layer. The SVM was chosen because it manages high-dimensional data effectively. We evaluated the model using 100, 50, and 20 principal components to assess the impact of dimensionality on performance. The table below summarizes the overall accuracy and class-wise precision, recall, and F1-scores for each configuration.

Number of PCs	Accuracy
100	0.9940
50	0.9930
20	0.9950

Table 4 SVM accuracy

Number of PCs	Class	Precision	Recall
100	Cat	0.99	1.00
100	Dog	1.00	0.99
50	Cat	0.99	1.00
50	Dog	1.00	0.99
20	Cat	0.99	1.00

Table 5 SVM precision/recall scores

As shown, the SVM performs very well, correctly classifying almost all samples in each class.

We also trained a linear layer on the original 2048-dimensional feature set. The model was trained for 20 epochs and reached stability around 3 epochs, as the loss plateaued afterwards [Figure 9]. The results shown below indicate robust performance, with high accuracy across classes.

Class	Precision	Recall	F1-score
0	1.00	0.99	0.99
1	0.99	1.00	0.99
Accuracy	-	-	0.99

Table 6 Linear Layer Classification

E. Clustering for Dataset (OxfordPetsIIT)

This dataset consisted of 37 classes with significant overlap, making clustering at the subclass level ineffective. Since each subclass belonged to either a dog or a cat, we mapped the labels to their respective super-classes, reducing the problem to 2 classes: Dog and Cat. Using the same approach as for the previous dataset, we applied K-Means clustering on feature vectors reduced to 100, 50, and 20 principal components. The clustering results are summarized below:

Dataset Size	clusters	Silhouette Score	Davies-Bouldin Index
100	2	0.300	2.668
50	2	0.313	2.228
20	2	0.362	1.819

The findings confirm the same result as with the other dataset, that reducing the dimensions significantly enhances cluster definition.

We can see a side by side comparison in [Figure 8]

F. Classification on Dataset (OxfordPetsIIT)

For classification, we evaluated two methods on the full set of thirty-seven classes: a support vector machine (SVM) and a linear layer as used in the previous dataset. We evaluated the model using 100, 50, and 20 principal components to assess the impact of dimensionality on performance. The table below summarizes the overall accuracy. Due to the substantial number of classes, class-wise F1, precision, and recall scores are not shown for clarity.

Number of PCs	Accuracy
100	0.87
50	0.88
20	0.85

Table 7 SVM accuracy

As can be inferred from the table, the accuracy remains relatively stable around 85 to 88 percent.

Similarly, we trained a linear layer with an input dimension of 2048 and an output dimension corresponding to the thirty-seven classes. The model was trained for 30 epochs and reached stability around 10 epochs, as the loss plateaued afterwards. On the holdout test set, the linear layer achieved an accuracy of **91%**, slightly higher than that of the SVM.

V. CONCLUSION

In this project, we implemented end-to-end machine learning pipelines for both low-dimensional tabular data and high-dimensional image data. This involved extensive preprocessing, including log-transformation and standardization for the climate dataset, and deep feature extraction using a pre-trained ResNet-50 architecture for the image datasets (OxfordPets and Cats vs Dogs). Principal Component Analysis (PCA) was rigorously investigated and applied in both tasks, proving essential for mitigating the curse of dimensionality and improving the performance of distance-based algorithms.

For the unsupervised learning task, we clustered the Basel Climate dataset using K-Means, Gaussian Mixture Models (GMM), and DBSCAN. After evaluating these using Silhouette Coefficients and the Davies-Bouldin Index, we selected K-Means (K=3) as the optimal approach, as it provided the most distinct separation of weather patterns compared to the density-based limitations of DBSCAN in the reduced feature space. For the image datasets, we demonstrated that clustering PCA-reduced deep features (20 components) could successfully recover class structures without labels.

Finally, we performed supervised classification on the extracted image embeddings. We compared the performance of a Support Vector Machine (SVM) against a custom Linear layer added onto the ResNet50 architecture. The Linear layer yielded superior accuracy, successfully capturing the non-linear relationships within the deep feature space that the linear models struggled to resolve.

VI. FUTURE WORK

In future work, the next step for this project would be to expand the comparative analysis to include modern architectures such as Vision Transformers (ViT) or EfficientNet to benchmark their feature quality against ResNet-50. It would also be valuable to implement "fine-tuning" of the neural network backbone rather than using it solely as a fixed feature extractor, which could potentially improve classification accuracy on fine-grained datasets like OxfordPets.

Secondly, the current analysis treats daily weather observations as independent samples, which represents a simplifying assumption. In reality, climate variables exhibit strong temporal dependencies, with weather conditions often persisting over multiple days. Future work could therefore incorporate time-series clustering approaches to better capture evolving weather regimes, for example by clustering sequences of observations rather than individual days. This would allow the model to account for temporal continuity and potentially identify patterns such as prolonged warm, cold, or wet periods that are not apparent under an independence assumption.

Given the computational complexity of extracting 2048-dimensional embeddings from the ResNet-50 backbone, a **stratified random sample** of the dataset was utilized for this experiment. This approach reduced the processing time to feasible limits for the available hardware while preserving the original class distribution to ensure the validity of the classification results.

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APPENDIX



Figure 6 PCA- Scree Plot

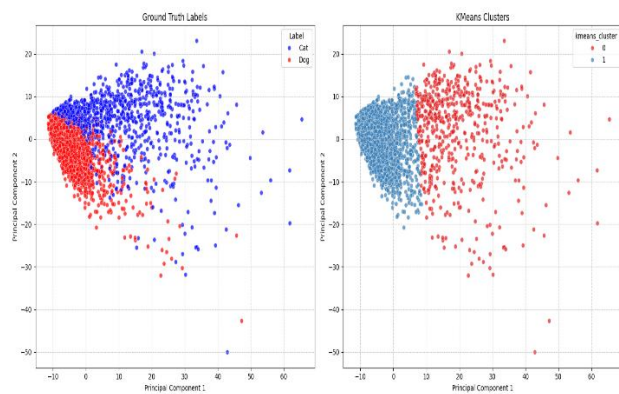


Figure 8 Original vs Predicted Labels

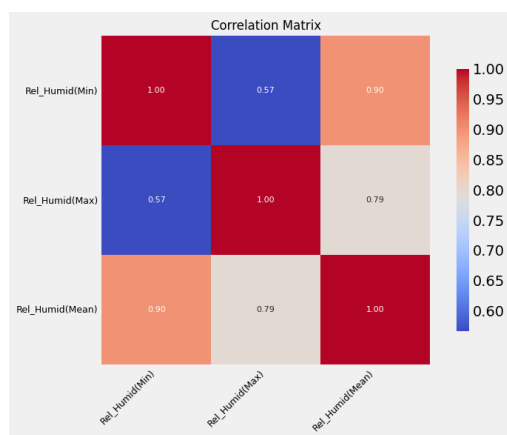


Figure 7 Collinearity Matrix

- [8] GeeksforGeeks (2020). *Residual Networks (ResNet) Deep Learning*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/deep-learning/residual-networks-resnet-deep-learning/>

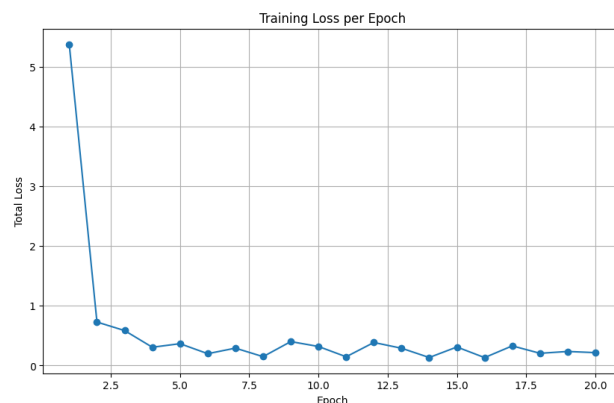


Figure 9 Loss Per Epoch